NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE | NTU Academy for Professional and Continuing Education

(SCTP) Advanced Professional Certificate

Data Science and AI

3.3 Supervised Learning Algorithms

Module Overview

3.1 Probability and Statistics for Machine Learning

3.2 Introduction to Machine Learning

3.3 Supervised Learning

3.4 Supervised Learning - Advanced

3.5 Unsupervised Learning

3.6 Time Series Data & Forecasting

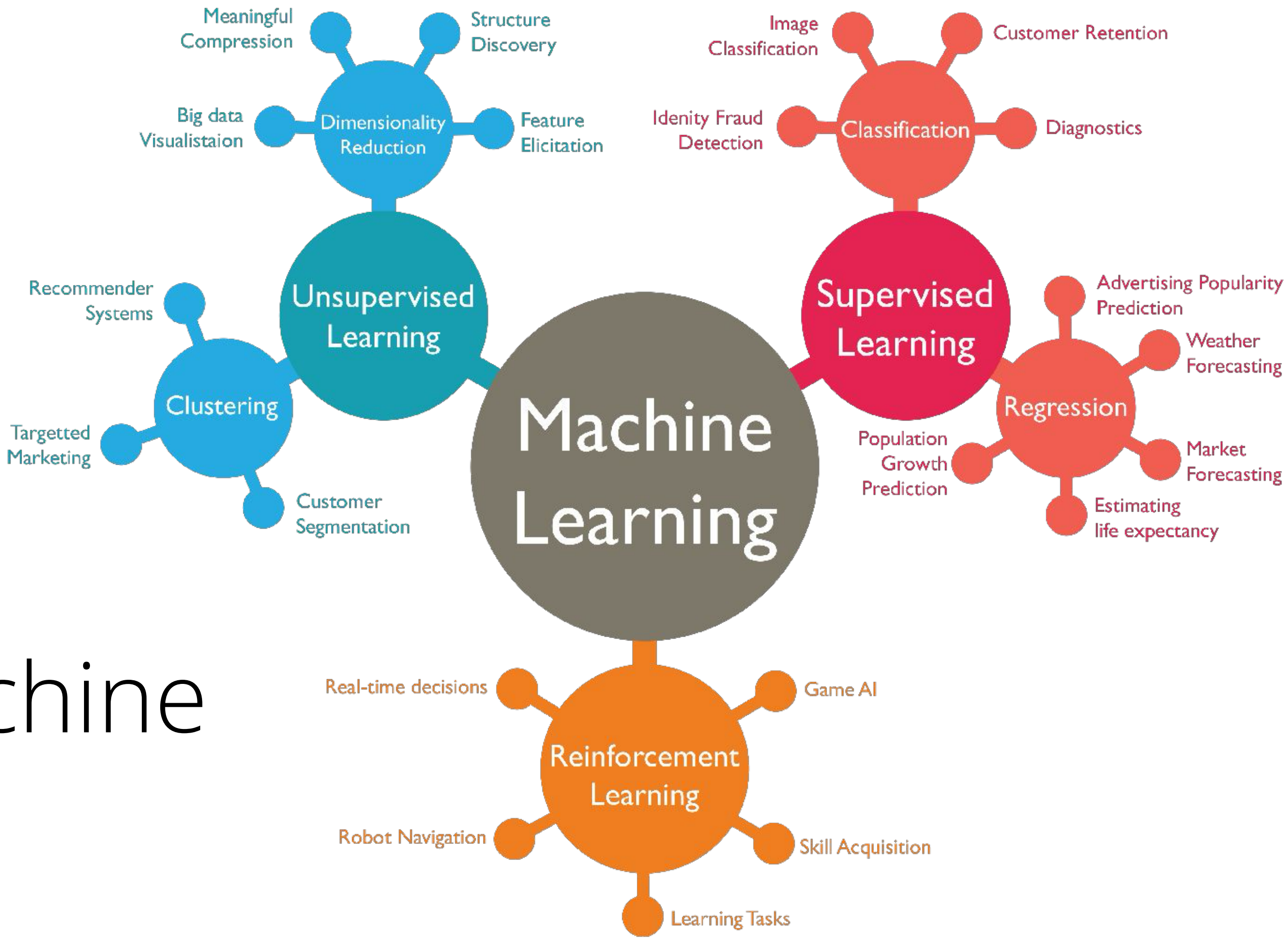
3.7 Neural Network & Deep Learning

3.8 Computer Vision (CV)

3.9 Natural Language Processing (NLP)

3.10 NLP - Advanced

Types of Machine Learning



Lesson Plan

1. Pre-processing numerical and categorical variables
2. Classification and regression metrics
3. End-to-end supervised learning workflow
4. Bias-variance trade-off

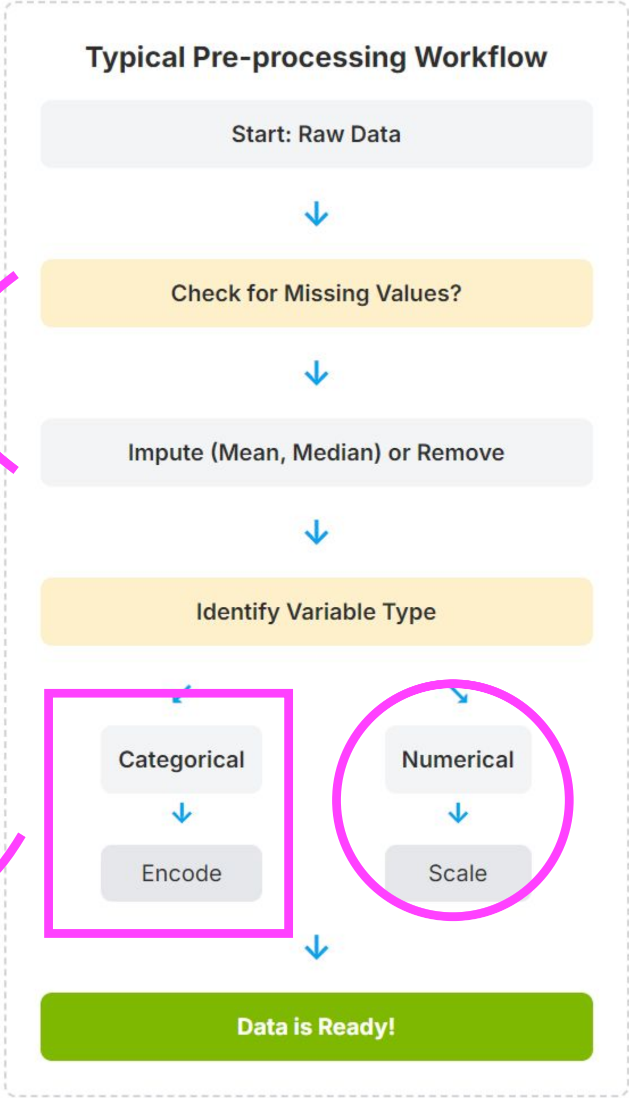
offline metrics

online metric

sklearn
pipeline api

eng metrics
latency
p99 ms

Pre-Processing Techniques



problem
formulation

EDA

preprocessing

e.g
5-room
4-room
landed
..

Pre-Processing Techniques

Categorical Variables

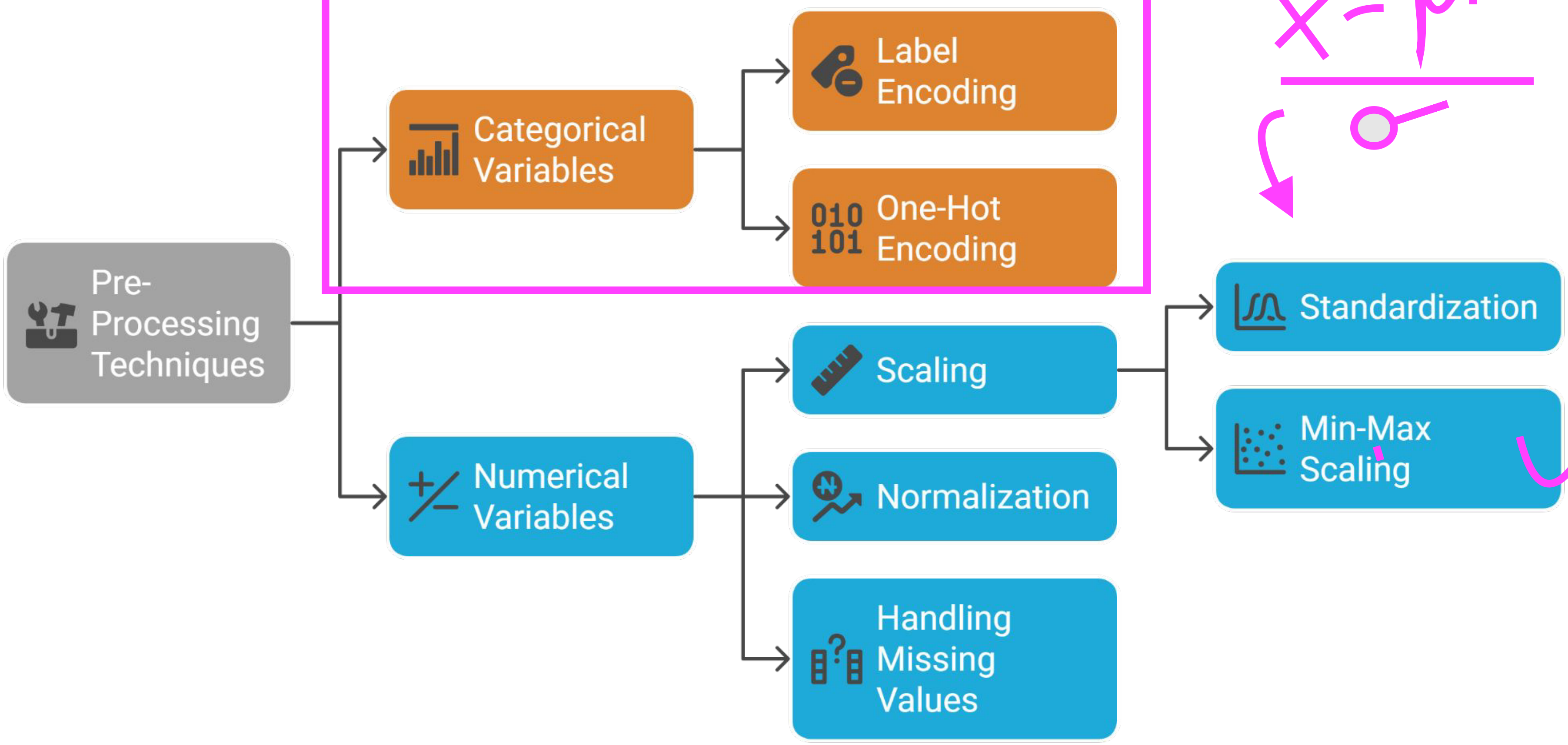
- Label encoding
- One-Hot encoding

M

Numerical Variables

- Scaling - standardization & min-max
- Normalization
- Handling missing value

Pre-Processing Techniques



$$\frac{x-\mu}{\sigma}$$

One-hot Encoding

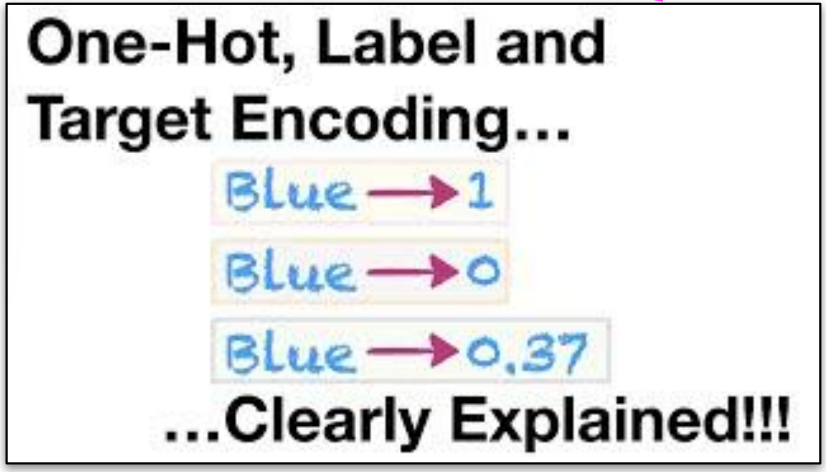
female=0 , ~~male~~=1

y

x

Converts categorical values into a binary vector where only one bit is set to 1 out of all bits

N - 1



Favorite Color	Height (m)	Loves Troll 2
Blue	1.77	Yes
Red	1.32	No
Green	1.81	Yes
Blue	1.56	No
Green	1.64	Yes
Green	1.61	No
Blue	1.73	No

cardinality

Blue	Red	Green	Height (m)	Loves Troll 2
1	0	0	1.77	1
0	1	0	1.32	0
0	0	1	1.81	1
1	0	0	1.56	0
0	0	1	1.64	1
0	0	1	1.61	0
1	0	0	1.73	0

27

https://youtu.be/589nCGeWG1w?t=84&si=b_yuWSpMpRxS-4zD (1:24 to 3:25)

One-hot Encoding

- One-hot encoding is best used for nominal data where no ordinal relationship exists.
- The downside is that it can lead to a **high-dimensional** feature space, which might be problematic for models that struggle with high dimensionality.

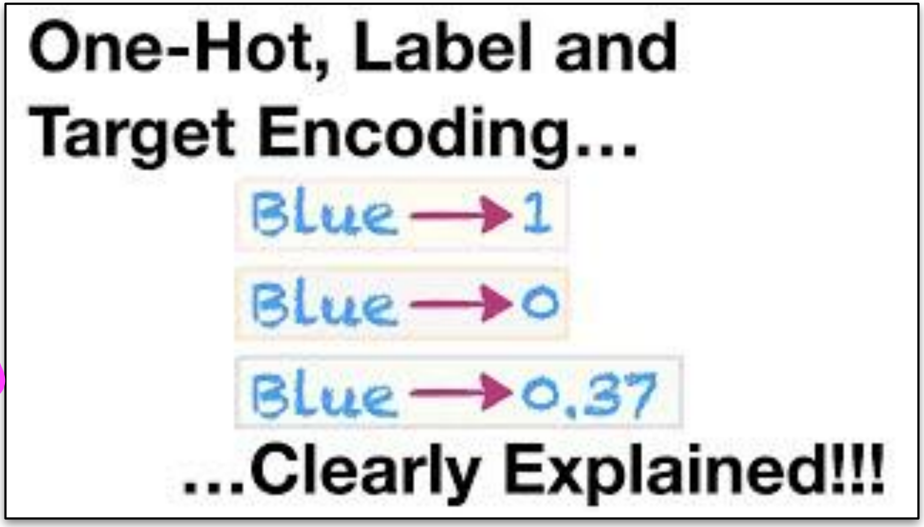
Label Encoding

Simple method where each unique category value is assigned an integer value

Favorite Color	Height (m)	Loves Troll 2
Blue	1.77	Yes
Red	1.32	No
Green	1.81	Yes
Blue	1.56	No
Green	1.64	Yes
Green	1.61	No
Blue	1.73	No

G > r > B

Favorite Color	Height (m)	Loves Troll 2
0	1.77	1
1	1.32	0
2	1.81	1
0	1.56	0
2	1.64	1
2	1.61	0
0	1.73	0



https://youtu.be/589nCGeWG1w?t=205&si=H9XSbS9OTbq2WVFY (3:25 to 4:49)

Label Encoding

- Label encoding is ideal for ordinal data, where the categories have some inherent order.
- However, using this method on nominal data (no intrinsic order) can introduce a **new problem**—the model might assume a natural ordering between categories which may result in poor performance or unexpected results.

Min-Max Scaling

- Finds the smallest and largest numbers in your set.
- Calculates a new value for each number so it fits within that 0 to 1 range.
- Ensures all your data points are on the same scale, which can be important for some machine learning algorithms.

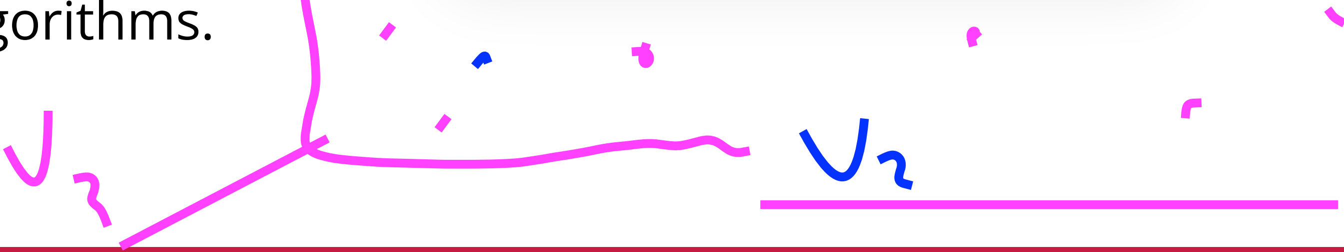
f_1, f_2

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

dollar $\boxed{\text{kg}}$ cm time

Min-max scaling rescales the feature to a fixed range, usually 0 to 1.

V_1



Standardization Scaling

x_i = [1,2,3,4,5]

x_1, x_10, x_1500

5

- Scaling adjusts the range of data so that different features contribute equally to the final prediction.

$$x' = \frac{x - \mu}{\sigma}$$

V_1

neural network

μ

σ

gradient descent
also requires

Standardization rescales data to have a mean (μ) of 0 and standard deviation (σ) of 1 (unit variance).

Train , Validation, Test

.fit
.transform

colour = [blue, green, red]

L2 Normalization

- Often used in text processing when unit vectors are required.

$$x' = \frac{x}{||x||_2}$$

L2 normalization scales input vectors individually to unit norm (vector length).

The Formula

For a vector $\mathbf{x} = [x_1, x_2, \ldots, x_n]$, the L2 norm is:

$$\|\mathbf{x}\|_2 = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$$

Let's take a simple vector $\boxed{\mathbf{x} = [3, 4]}$ and apply L2 Normalization to find $\mathbf{x}'$.

1. Calculate the L2 Norm ($\|\mathbf{x}\|_2$):

$$\sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = \mathbf{5}$$

2. Divide the original vector by the norm:

$$\mathbf{x}' = \frac{[3, 4]}{5} = \boxed{[\mathbf{0.6}, \mathbf{0.8}]}$$

Handling Missing Values

Missing values can significantly affect the performance of machine learning models. Common strategies for handling missing data include imputation and removing records with missing values.

- **Imputation**

  Fills in missing values with a specific value, such as the mean, median, or mode of the column.

- **Removing Missing Values**

  If the dataset has only a few missing values, it might be reasonable to drop those records. However, this can lead to loss of valuable data.

Classification Metrics

$$\underline{f(x)} \to \hat{y} = 0 \sim 1 \to [0, 1]$$

Classification Metrics

- Confusion Matrix
- Accuracy
- Precision
- Recall
- Specificity
- F1 Score

$y = \bar{1}, 0$

$\hat{y} = [1, 0]$

Confusion Matrix

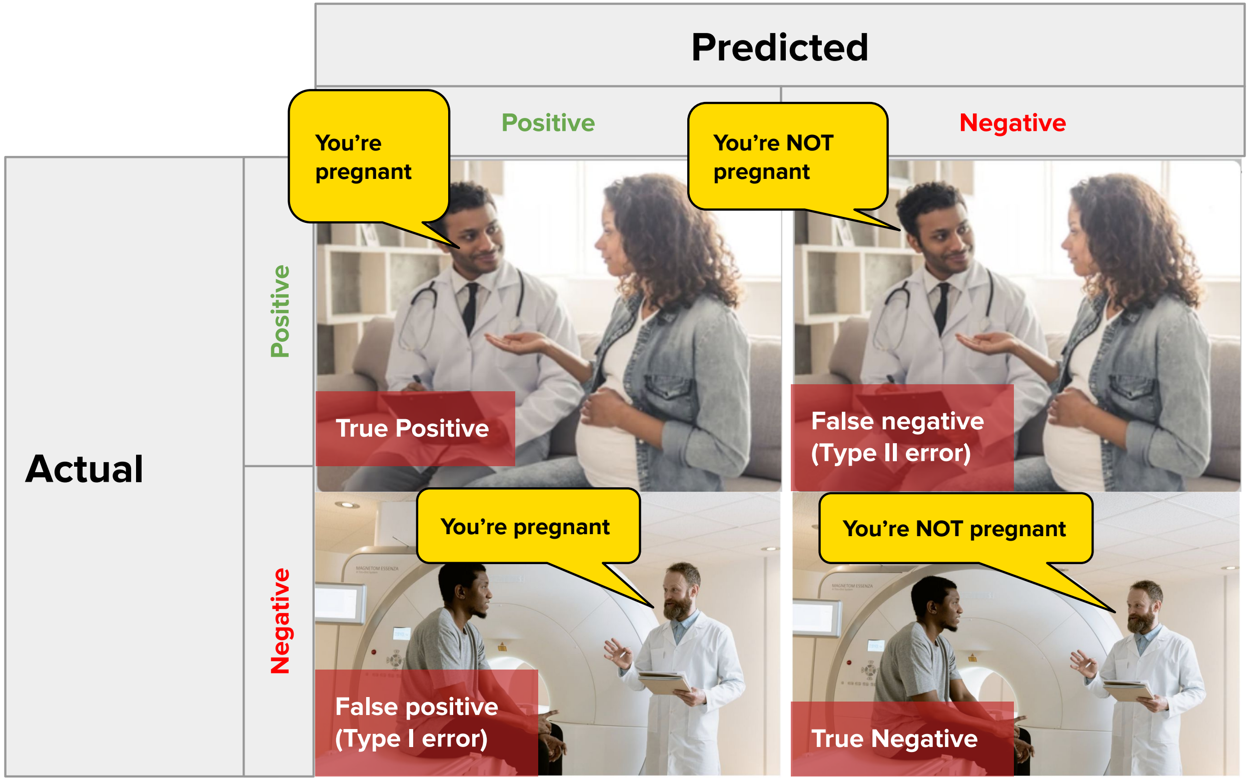
2*2

- **True Positive (TP)**:
  Correctly predicted positives
- **True Negative (TN)**:
  Correctly predicted negatives
- **False Positive (FP)**:
  Incorrectly predicted positives
  (*Type I error*)
- **False Negative (FN)**:
  Incorrectly predicted negatives
  (*Type II error*)

		Predicted	
		Negative (N) -	Positive (P) +
Actual	Negative -	True Negative (TN)	**False Positive (FP) Type I Error**
	Positive +	**False Negative (FN) Type II Error**	True Positive (TP)

Table that is used to describe the performance of a classification model on a set of test data for which the true values are known.

Remembering the confusion matrix



*Source: Confusion Matrix memes

Accuracy

TP+TN / (TP+FP+TN+FN)

Accuracy is best used when the target classes are well balanced.

However, it can be misleading when dealing with imbalanced datasets.

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Observations}}$$

The ratio of correctly predicted observations to the total observations.

Transaction	Actual Category (y)	Predicted Category (y_hat)
Transaction 1	Normal	Normal
Transaction 2	Normal	Normal
Transaction 3	Normal	Normal
Transaction 4	Normal	Normal
Transaction 5	Normal	Normal
Transaction 6	Normal	Normal
Transaction 7	Normal	Normal
Transaction 8	Normal	Normal
Transaction 9	Fraud	Normal
Transaction 10	Fraud	Normal

Precision

$\hat{y} = 1, 1$

Use precision when the cost of a false positive is high, such as in spam email detection.

*"Of the items I predicted as **positive**, how many were **actually** positive?"*

$$\text{Precision} = \frac{\text{TP}}{\boxed{\text{TP} + \text{FP}}}$$

Ratio of correctly predicted positive observations to the total predicted positive observations.

Recall (Sensitivity)

$y = 1$, $\hat{y} = 1$

- Also known as True Positive (TP) Rate
- Useful when **missing positives is risky**
- ***"Of all actual positives, how many did I catch?"***

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

Ratio of correctly predicted positive observations to all observations in the actual class.

$\hat{y} = \text{Prob} \rightarrow [1, 0]$

0   1

Specificity

Also known as True Negative (TN) Rate

- It complements recall (sensitivity) by focusing on the model's performance with the negative class.
- *"Of all actual negatives, how many did I correctly say were negative?"*
- E.g. in medical diagnostics, a false positive might lead to unnecessary treatment, which could be costly or harmful.

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}}$$

Measures the proportion of actual negatives that are correctly identified negative.

F1 Score

P R

- This score takes both false positives and false negatives into account.
- Use the F1 score when you want to balance precision and recall, especially if there is an uneven class distribution.

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Harmonic mean of **Precision** and **Recall**.

(x_1 + x_2+ x_n) * (1/N)

Summary of Classification Metrics

Metric	Usage Scenario
Accuracy 🎯	**Gender Classification:** When a model decides if gender is Male or Female. There is roughly equal counts of each gender, and misidentifying one is no worse than the other.
Precision 📌	**Legal Sentencing / Death Penalty:** The principle "Better that ten guilty persons escape than that one innocent suffer." A False Positive (convicting the innocent) is the ultimate failure.
Recall / Sensitivity 📝	**Home Smoke Alarms:** It is okay if the alarm chirps when you burn toast (False Positive), but it is a disaster if it stays silent during a real fire (False Negative).
Specificity 👓	**Drug Testing for the Olympics:** You must ensure that a clean athlete is never labeled as a "doper." The priority is protecting the reputation of the innocent/negative cases.
F1 Score 🏎️	**Credit Card Fraud Detection:** Only 0.01% of transactions are fraud. Accuracy is useless here. F1 forces the model to find fraud (Recall) without flagging every single honest purchase (Precision).

F_β

Summary of Classification Metrics

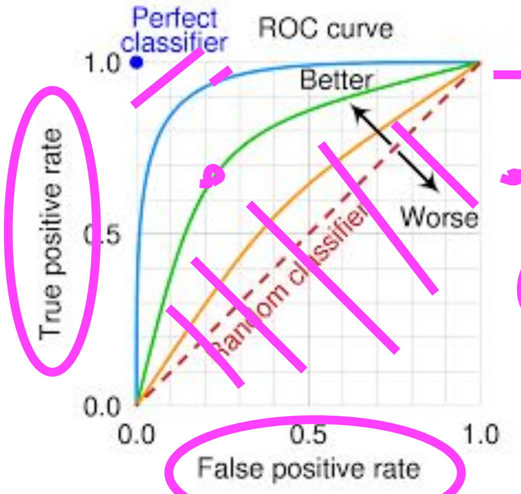
Metric	Definition	Primary Application	Limitations
Accuracy	Proportion of total predictions that were classified correctly.	Best for balanced datasets where all classification error types have equivalent costs.	Can be misleading on imbalanced datasets, as a model might perform well by only predicting the majority class.
Precision	Proportion of positive identifications that were actually correct.	Critical when the cost of a false positive is substantial.	Optimizing for precision can decrease **recall**, as a model might become overly conservative and miss many true positive cases.
Recall / Sensitivity	Proportion of actual positives that were correctly classified.	Essential when the cost of a false negative is high, such as in medical diagnostics.	Maximizing recall can inadvertently reduce **precision** by generating a higher number of false positives.
Specificity	Proportion of actual negatives that were correctly classified.	Relevant when false positive costs are high and the main objective is to confirm negative cases.	High specificity can be accompanied by low **recall**, meaning a focus on identifying negatives may lead to missing positive instances.
F1 Score	The harmonic mean of precision and recall, providing a single metric that balances the two.	Good for imbalanced datasets where balancing precision and recall is necessary due to significant costs for both error types.	Can be less intuitive than individual metrics and assumes **precision** and **recall** are equally important.

Summary of Classification Metrics

Metric	Definition	Primary Application	Limitations
Accuracy 🎯	$\frac{\text{Number of Correct Predictions}}{\text{Total Number of Observations}}$	Best for balanced datasets where all classification error types have equivalent costs.	Can be misleading on imbalanced datasets, as a model might perform well by only predicting the majority class.
Precision 📌	$\frac{TP}{TP + FP}$	Critical when the cost of a false positive is substantial.	Optimizing for precision can decrease **recall**, as a model might become overly conservative and miss many true positive cases.
Recall / Sensitivity 📝	$\frac{TP}{TP + FN}$	Essential when the cost of a false negative is high, such as in medical diagnostics.	Maximizing recall can inadvertently reduce **precision** by generating a higher number of false positives.
Specificity 👓	$\frac{TN}{TN + FP}$	Relevant when false positive costs are high and the main objective is to confirm negative cases.	High specificity can be accompanied by low **recall**, meaning a focus on identifying negatives may lead to missing positive instances.
F1 Score 🏎️	$2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$	Good for imbalanced datasets where balancing precision and recall is necessary due to significant costs for both error types.	Can be less intuitive than individual metrics and assumes **precision** and **recall** are equally important.

Receiver Operating Characteristic (ROC) Curve

- For evaluating performance of classification models, particularly in binary classification.
- Visualize and quantify trade-offs between TP rate (**sensitivity**) and FP rate (**1 - specificity**).
- Created by plotting the TPR against the FPR at different threshold values.
- A threshold is a point above which a given observation is classified as belonging to the positive class.

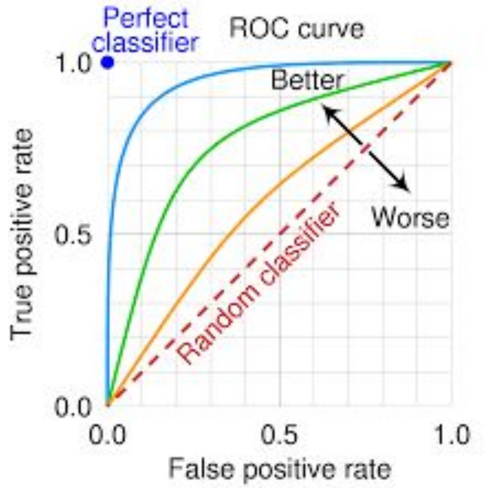


0.5

0 — 1

Interpretation of the ROC Curve

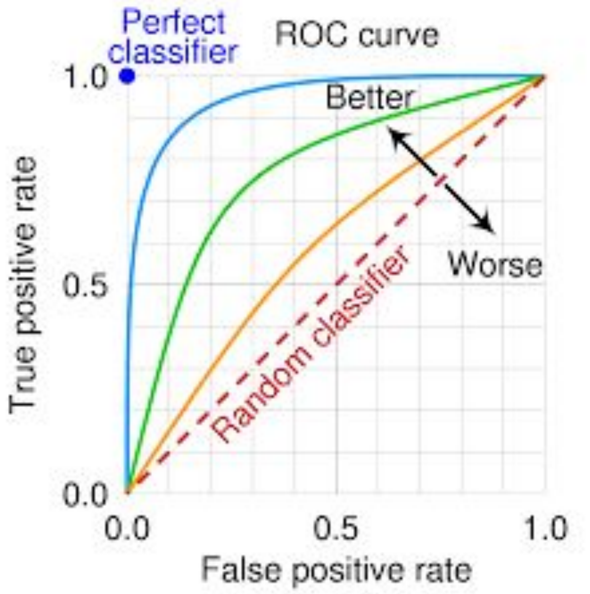
- A curve near the top-left corner indicates a good performance.
- The diagonal line (FPR = TPR) represents random guessing.
- The area under the diagonal is 0.5, which is the AUC for a random classifier.



Area Under Curve (AUC)

The Area Under the Curve (AUC) represents the degree of separability between the classes. Used as a summary of the ROC curve.

- **AUC = 1**: Perfect classifier. All positive instances rank higher than all negative instances.
- **0.5 < AUC < 1**: Good classifier. Higher AUC indicates better performance.
- **AUC = 0.5**: Random classifier. No discriminative power.
- **AUC < 0.5**: Worse than random guessing, but by inverting the predictions, it could be useful.



Usage

y = 0 vs 1

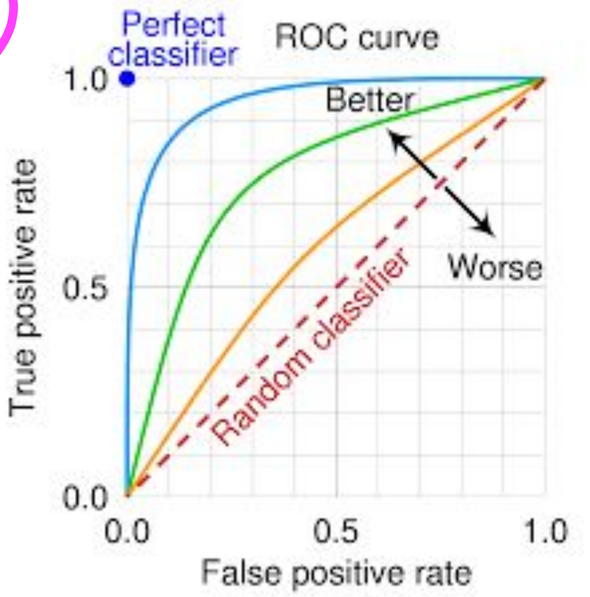
Invariance to Class Distribution:

ROC and AUC are useful especially when dealing with imbalanced classes.

Threshold Independence:

They measure the quality of the model's predictions without tying to a specific threshold.

ROC and AUC are useful when you need to evaluate a model's performance across different classification thresholds.



$y, \hat{y} \rightarrow$ continuous value

Regression Metrics

$\boxed{OLS}$

error

Regression Metrics

$n \quad \underbrace{\frac{1}{N}\sum(y-\hat{y})^2}$ $\boxed{SSR}$

- 1 Mean Squared Error (**MSE**) is the average of the squared differences between the predicted values and the actual values. It gives a higher weight to larger errors.
- 4 Mean Absolute Error (**MAE**) is the average of the absolute differences between the predicted values and the actual values. It gives an idea of how wrong the predictions were.
- 2 Root Mean Squared Error (**RMSE**) is the square root of the mean squared error. It is one of the most widely used metrics for regression tasks.
- 3 **R-squared** is a statistical measure that represents the proportion of the variance for the dependent variable that's explained by the independent variables in a regression model.
- **Adjusted R-squared** is a modified version of R-squared that has been adjusted for the number of predictors in the model.

LR

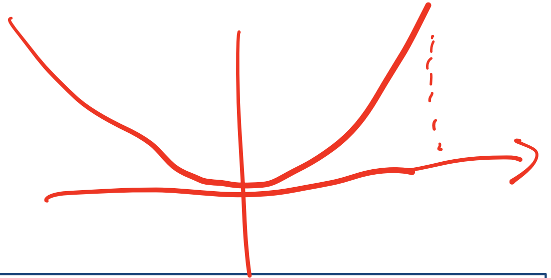
Mean Absolute Error (MAE)

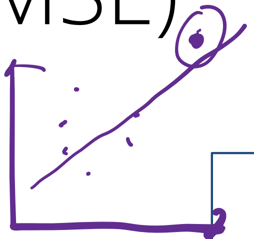
- Gives an idea of how wrong the predictions were.
- Measures the average magnitude of errors in a set of predictions, without considering their direction.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

MAE is the average of the absolute differences between the predicted values and the actual values.

Mean Squared Error (MSE)





error

$y - \hat{y}$

- MSE is sensitive to outliers.
- Gives a higher weight to larger errors.
- Larger errors have a disproportionately large effect on MSE, making it useful when large errors are particularly undesirable.

10 sample

2 sample

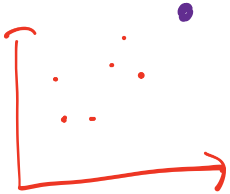
$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

$2 \rightarrow 4$

$10 \rightarrow 100$

30,000 000

MSE is the average of the squared differences between the predicted values and the actual values.



OLS → SSR → MSE

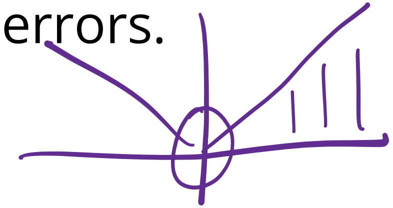
Root Mean Squared Error (RMSE)

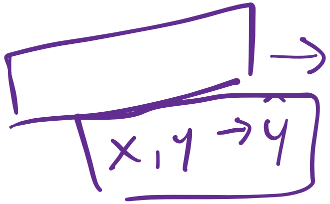
- RMSE is the square root of the mean squared error. It is one of the most widely used metrics for regression tasks.
- RMSE is in the same units as the response variable.
- Like MSE, it gives a higher weight to larger errors.

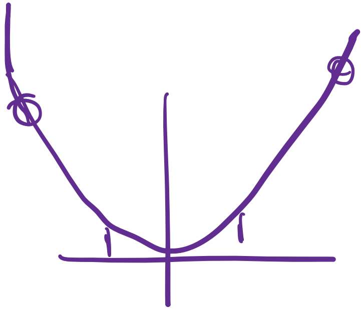
$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

MSE

LR







Metric	Interpretation	Advantages	Disadvantages
Mean Absolute Error (MAE)	Lower is better. Represents average absolute difference between predicted and actual values.	Same unit as target variable. Directly interpretable	Less sensitive to large errors. Derivatives not defined at zero
Mean Squared Error (MSE)	Lower is better. Represents average squared difference between predicted and actual values.	Penalizes larger errors more heavily. Mathematically convenient for optimization (Differentiable)	Not in same units as target variable. Hard to interpret in absolute terms
Root Mean Squared Error (RMSE)	Lower is better. Square root of MSE, bringing it back to the same units as the target variable.	Same unit as target variable. Penalizes large errors. Popular in scientific literature	Still sensitive to outliers. Less interpretable than MAE
R-squared (R^2)	Higher is better. Proportion of variance explained by the model.	Scale-invariant (0-1 range). Comparable across datasets	Can increase with irrelevant features. Doesn't indicate prediction accuracy
Adjusted R-squared	Higher is better. R^2 adjusted for number of predictors.	Penalizes excess features	Still doesn't indicate prediction accuracy. Can be misleading for non-linear relationships

model

offline metric

online metric

minimize loss

ROC-AUC-

CTR

Part 1: Preprocessing Categorical & Numerical Values

loss function

Part 2: Regression Metrics

Part 3: End-to-end Workflow on Titanic Data Set

End-to-End Workflow

Apply to Titanic dataset to predict survival rate of passengers.

$\boxed{\text{Bias-Variance}}$ Trade-off in ML

unfit

overfit

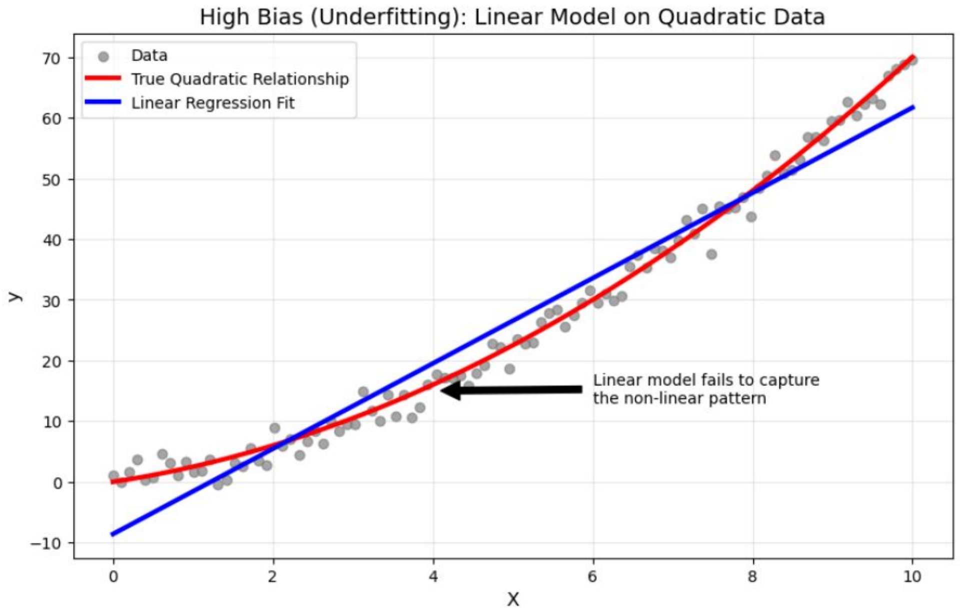
Bias

variance

What is Bias?

underfit

- Bias refers to the error that is introduced by approximating a *complex* real-world problem, by a *simplified* model.
- Measures the difference between the expected prediction of our model and the true values.
- High bias can cause the model to miss relevant relations between features and target outputs (also known as *underfitting*).

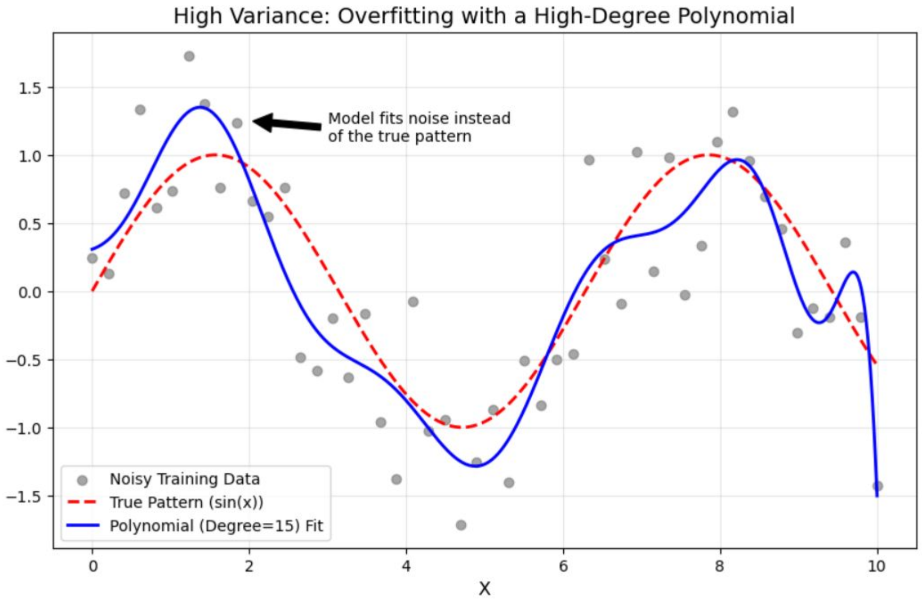


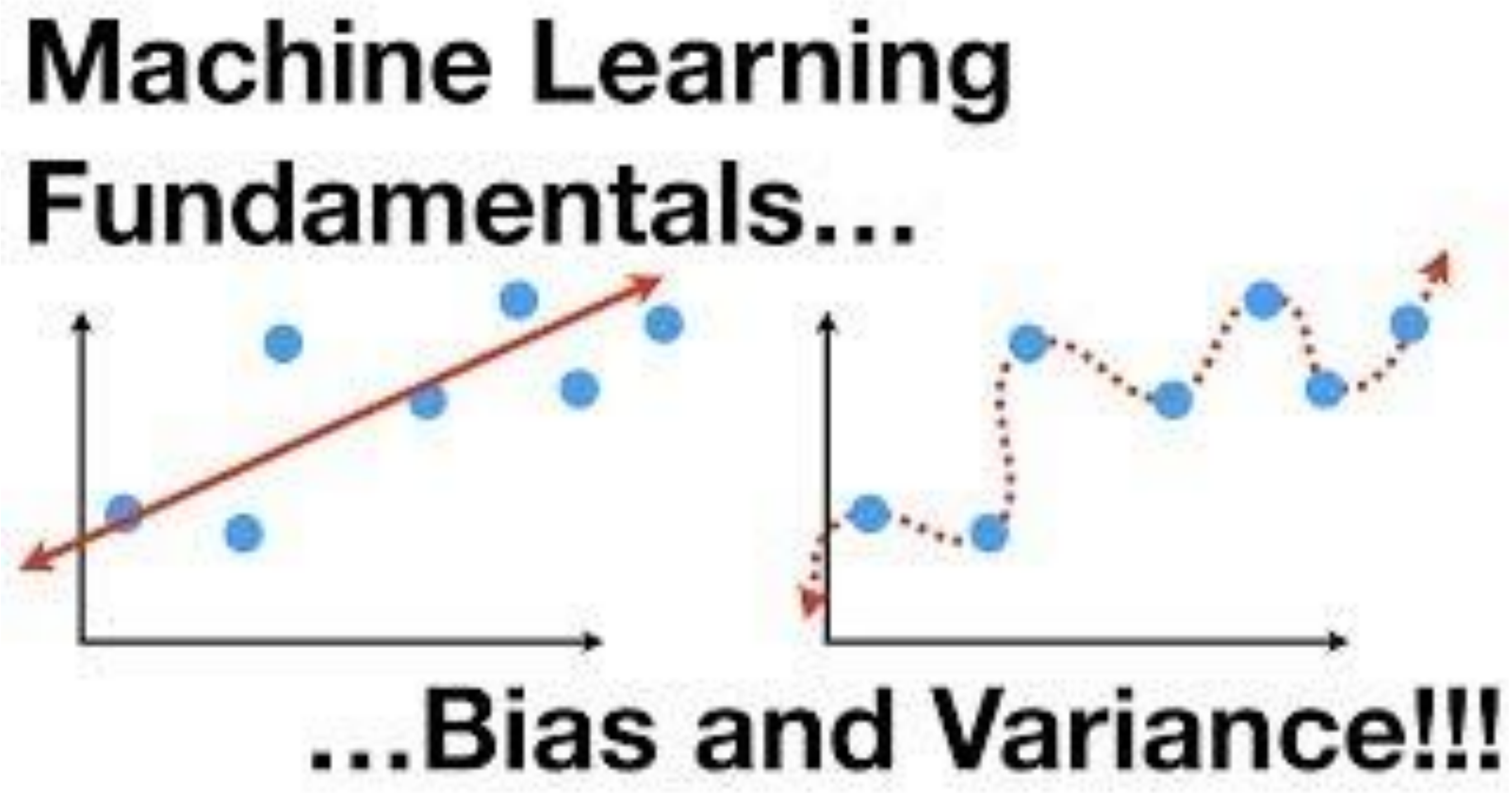
What is Variance?

$f(x) \rightarrow \hat{y}$

$f(x_2) \rightarrow \hat{y}_2$

- Variance refers to the error that is introduced by the model's sensitivity to the fluctuations in the training dataset.
- A model with high variance pays a lot of attention to training data and does not generalize on the data which it hasn't seen before (also known as *overfitting*).





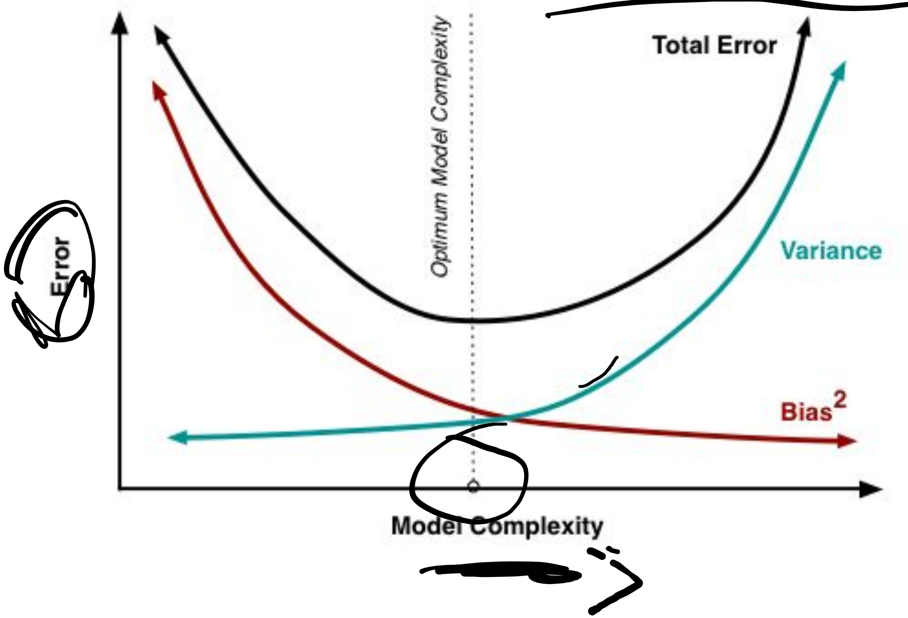
https://www.youtube.com/watch?v=EuBBz3bI-aA

The Tradeoff

$y = \beta_0 + \beta X$ → vs $y = \beta_6 + x_1^2 + x_2 + x_{\ldots}$

The bias-variance tradeoff is the balance between the error introduced by the bias and the variance.

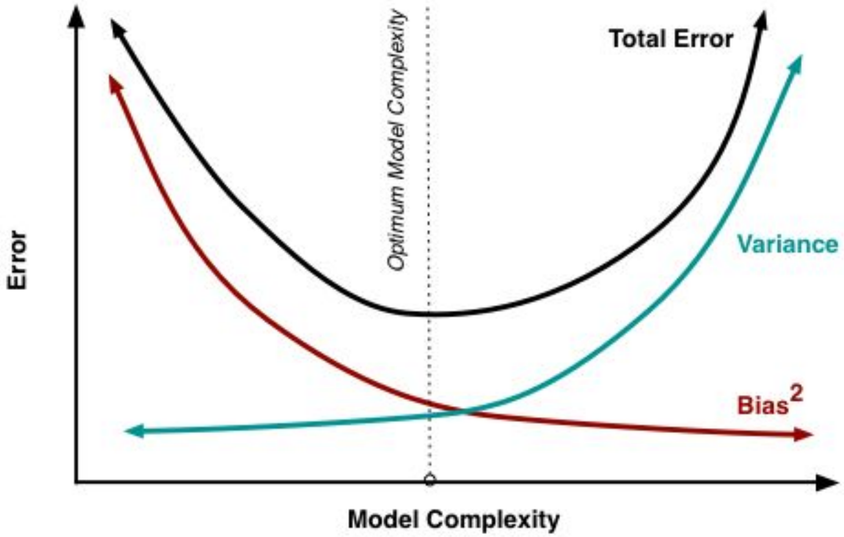
Ideally, we want to choose a model where both bias and variance are as low as possible.



The Tradeoff

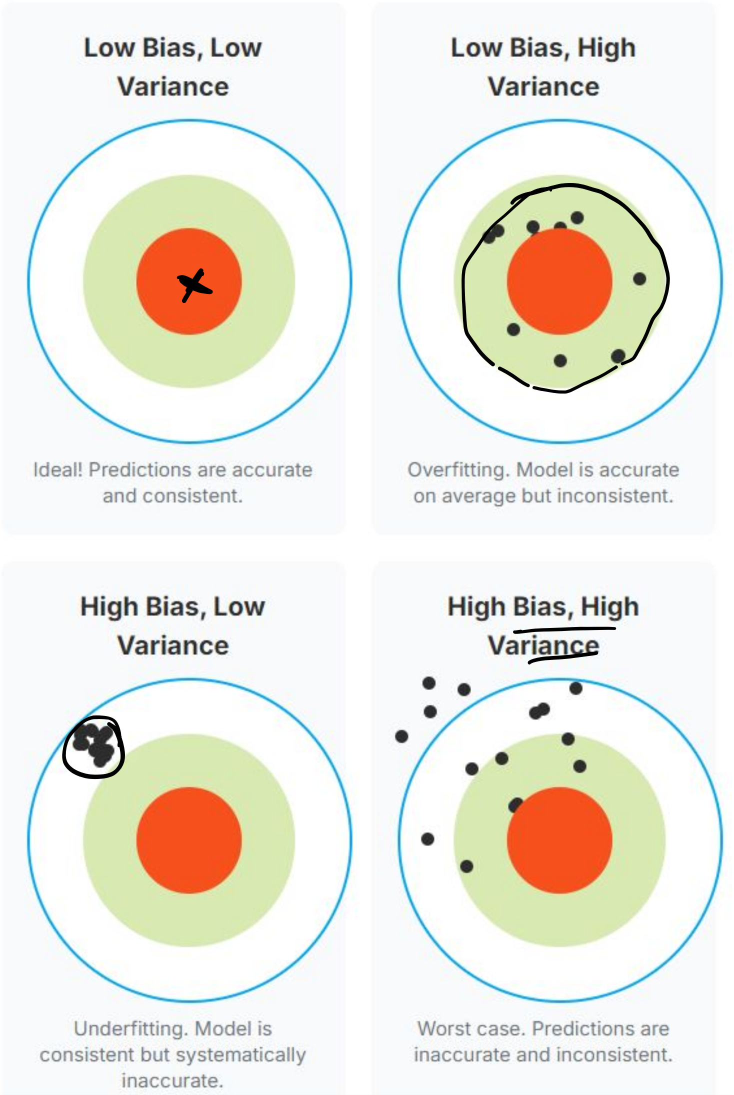
In practice, decreasing one will often increase the other.

- A model with high bias (simple models) will not be complex enough to capture the patterns in the data, leading to underfitting.
- Conversely, a model with high variance (complex models) will capture noise in the training data, leading to overfitting.



The Tradeoff

- High bias is when our predictions are consistently far off from the target.
- High variance is when our predictions are all over the place around the target.
- Low bias and low variance are when our predictions are consistently close to the target.



Balancing Bias and Variance

200

77

To achieve a model that generalizes well, we need to find a sweet spot that minimizes both bias and variance.

Cross-validation: Helps in assessing how the results of a statistical analysis will generalize to an independent dataset.

Training with more data: Helps to reduce variance without increasing bias.

Feature selection: Removing irrelevant features can reduce variance without increasing bias too much.

Regularization: Techniques like Lasso and Ridge can reduce variance at the cost of introducing some bias.

Ensemble methods: Combining predictions from several models can reduce both bias and variance.

Part 4: Bias-Variance Trade-offs in ML

End of Lesson - Exit Ticket

Survey Link

Slido link will be provided in class